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Neurosurgeon Survival Strategy on Saturn during Healthcare Reform

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ABSTRACT / 摘要 Medical practice on Saturn, a socialist planet with a “Medicaid for All” healthcare system, presents unique economic and ethical challenges. Over the past decade, Saturn’s government has transitioned from a free-market model to a fee-for-service system, and subsequently to a diagnosis-intervention packet system, followed by the introduction of a "work points system." These reforms aim to further limit reimbursements for surgeries and reduce medical costs, which may create significant survival pressures for physicians. Neurosurgery is one of the most complex and dedicated fields in medicine. Survival stress may compel neurosurgeons to engage in various unethical practices. This paper introduces the survival strategies of two neurosurgeons, Dr. Fisher and Dr. Sand, during these significant healthcare reforms on Saturn.

INDEX TERMS / 关键词 Healthcare reform; socialism; neurosurgery; ethics; physician salary

IMPACT STATEMENT / 影响声明 I certify that there is no actual or potential conflict of interest in relation to this article.

I. INTRODUCTION

Neurosurgical practice on Saturn has been documented (1). It differs significantly from that on Earth due to Saturn's limited resources, dense population, and harsh living conditions. To maximize survival chances, humans on Saturn have established a socialist society with a “Medicaid for All” healthcare system.

As Saturn develops, healthcare spending has risen sharply, placing significant pressure on the government's budget. In response, the government has conducted significant healthcare reform in the last decade, transitioning from a fee-for-service payment model to diagnosis-intervention packet (DIP) systems, annual salary,

and work point systems to control costs. Neurosurgery requires highly trained staff and advanced technology, which are particularly costly. In economically disadvantaged countries, neurosurgeons face various economic and ethical challenges. This study explores two neurosurgeons' survival strategies during the dramatic healthcare reform in Saturn.

II. Result

1. Living with kickback

In August, TO18 Saturn year, an early 60s patient presented with sudden onset of coma lasting for 5 hours was admitted to the neurosurgery center, a top-tier hospitals in the Capital of Saturn. Upon physical examination, the patient's Glasgow Coma Scale (GCS) score was 5, with bilateral pupils reactive to light and measuring 2 mm on both sides. A CT scan revealed a 2x2 cm hematoma in the brainstem. Given the patient's history of hypertension, hypertensive hemorrhage was the most likely diagnosis.

Dr. Fisher (pseudonym), the neurosurgeon on call, discussed the situation with the patient's daughter. Given the size of the hematoma, conservative management was considered appropriate. However, in the public hospital of Saturn, neurosurgeons are significantly underpaid. Dr. Fisher can only earn 100,000 Satolar per year. His son hopes to attend a prestigious local school, owning a house in the area was mandatory. The average house price was 2,000,000 Satolar, making it nearly impossible for a neurosurgeon to afford.

"Luckly", a pharmaceutical company offered Dr. Fisher a 2,000 Satolar "kickback" for each surgery using their products. Faced with financial pressures, Dr. Fisher felt there was little reason not to proceed with surgery. Although hematoma aspiration was a viable option, Dr. Fisher had to perform a craniotomy to use the company's products, including artificial dura and cranial bone fixation. The surgery was successful, but the patient remained in a coma despite six months rehabilitation.

2. Pay to have a job

Saturn government was determined to put an end to all medical kickbacks in public hospitals. The

DIP system was introduced to prevent excessive reimbursement claims by doctors. The DIP system places a cap on the costs associated with each disease treated in the hospital.

In May, TO24 Saturn year, Dr. Fisher was called to the emergency room (ER) for evaluating a traumatic patient. A high-speed motorcycle collided with a pedestrian. The pedestrian, a 20s male was found unconscious with an open injury in his front head. In physical examination, the patient has a GCS score of 3, his bilateral pupils dilated to 5 mm, and non-reactive to light. CT imaging revealed a large left-frontotemporal epidural hematoma and a 10 mm contralateral subdural hematoma, with diffuse cerebral edema. At the same time, an early 60s male patient arrived in the ER, complaining of severe headache and left-sided hemiparalysis. The patient narcolepsy, with bilateral pupils measuring 2 mm and reactive to light. His left side exhibited grade 2 hemiparesis. Head CT revealed a 1x1 cm right thalamic hemorrhage extending into the bilateral ventricles.

There were only two neurosurgeons on duty: Dr. Fisher and a resident, Dr. Sand (pseudonym), who had recently obtained medical approval to perform neurosurgery. According to local guidelines, the pedestrian did not meet the criteria for surgery due to his poor prognosis. However, in case of a car accident, the patient's medical expenses would be covered by the car owner's commercial insurance. This type of case was considered "ideal patient" under the DIP system, as the treatment cost would not impact the doctor's salary, regardless of the amount spent. Given this, Dr. Fisher decided to document the patient's physical examination as "GCS 4, bilateral pupil dilation", to justify his decision for surgery. He instructed Dr. Sand to manage the later patient with thalamic hemorrhage alone. Thalamic hemorrhage is considered a "bad disease" in their center, because the DIP system assigns a relatively lower reimbursement for this diagnosis. Treating a thalamic hemorrhage patient very likely result in financial penalties for exceeding the DIP cost limit. As a young surgeon, Dr. Sand was willing to gain surgical

experience at the cost of money, believing it would allow him to advance to an attending position more quickly. Therefore, he did not refuse Dr. Fisher's request.

Dr. Sand performed an endoscopic Transcortical-Transventricular approach for hematoma evacuation. Due to the presence of diffuse bilateral ventricular hematoma, an external ventricular drain was required. To minimize the surgery cost, he opted for a cheaper drainage tube without anti-reflux feature. The ventricular hematoma was successfully evacuated after one week of drainage; however, the patient complicated with intracranial infection. Dr. Sand referred the patient to a secondary hospital, to ensure that the additional medication costs would not be charged to his account. The patient was successfully treated with intravenous vancomycin and meropenem for another two weeks and ultimately recovered.

3. Fee-for-service, flee from service

In the year TO28, the DIP reform spread widely across Saturn. Surprisingly, it did not reduce expenditure. The unintended consequence of this policy was that neurosurgeon, incentivized by cost-efficiency, became increasingly focused on treating conditions deemed to have the highest “cost-benefit ratio”, such as asymptomatic arachnoid cysts, incidentalomas, and benign skull tumor. These conditions were technically simple, low-risk, and relatively high-cost. As a result, the surgical indications were frequently expanded, leading to an increase in number of surgeries performed.

Two years later, the introduction of the “annual salary system” was invented to address this issue. Attending neurosurgeons were limited to have a maximal 150,000 Satolar annually, while their performance was judged by the “work points system”. Surgeons are assigned specific "points" for each surgery they perform. Their final salary fluctuates depending on how much points they earned.

Dr. Fisher has been promoted to consultant. Now he is a famous surgeon and practices in both public and private hospitals. He demonstrated a clear preference for working in private hospitals where he can earn extra reimbursement besides

his annual salary. In June A TO29, an early 50s female patient presented to his public hospital clinic with a two-month history of recurrent headaches, dizziness, and nausea. Neurological examinations revealed no focal deficits. The head CT angiography identified a 12-mm saccular aneurysm at the bifurcation of the right middle cerebral artery, which showed signs of impending rupture.

Endovascular embolization was the treatment of choice. But instead of administrating the patient in the public hospital, Dr. Fisher referred the patient to a private hospital for the procedure, citing that “the long waiting times in the public system”. The truth is that, because Saturn government cut all the kickback in public hospital by “centralized procurement” (a similar policy conducted by China(2)), he can only receive kickback from private one. During the embolization, although the aneurysm can be filled with two large coils, he put eight smaller coils to achieve a similar packing density. This ultimately led to a 200% increase in costs. While the fee at private hospital was not covered by insurance, Dr. Fisher had successfully saved money for public healthcare.

Dr. Sand is now an attending physician. Today, a late 60s male was presented to his clinic, with complaints of progressive tremors in his right hand and difficulty walking. The patients had been diagnosed with Parkinson’s disease 3 years ago. He had been managed on a combination of levodopa and dopamine agonists, but symptoms were no longer well controlled. Dr. Sand find that the patient was of surgical indication for deep brain stimulation (DBS). But with the advent of the Work Points System, the reimbursement for complex surgeries, such as for Parkinson’s disease, was minimal. A DBS procedure was allocated only 76 points, even less than the reimbursement for a simple cecectomy (82 points). Another option is to refer the patient to the private secondary hospital for Gamma Knife treatment. Dr. Sand can earn 5,000 Satolar for each patient he referred to, with no risk of performing surgery himself.

Dr. Sand decides to "sell" the patient. He knows giving up surgery is digging his own career’s

grave. But surviving on Saturn is his main priority. The private hospital makes money, the patient gains access to more advanced treatments (Gamma Knife is not available in the Saturn public hospital), and it seems like a win-win-win for the patient, the private hospital, and himself. So, why not?

III. Discussion

Throughout human history, no healthcare reform has ever managed to satisfy all stakeholders. The primary goals of "Medicaid for All" have consistently been to expand coverage and reduce costs. Saturn exemplifies a successful healthcare reform, albeit with a significant financial shortfall, that seeks to achieve broader coverage by cutting costs and lowering service quality.

In labor-intensive countries, reducing labor costs often serves as the starting point for many reforms. This issue is particularly acute in low-income regions like Saturn, where neurosurgeons, already earning low hourly wages, may feel pressured to compromise the quality of their work in order to secure more time for rest. Without a strong ethical foundation and sense of responsibility, it becomes challenging for surgeons to dedicate the necessary time and effort to maintain even the most minor aspects of a patient's well-being. Ultimately, a surgeon's priority often shifts to mere survival.

Here, based on two neurosurgeon's experience, we introduce the survival strategy for neurosurgeon in Saturn during healthcare reform.

1. Make Every Surgery Bankable

Although neurosurgery is one of the most delicate fields in medicine, it does not necessarily guarantee a high income. In fee-for-service systems like that of Saturn, neurosurgery requires significant time and expensive equipment, often making it less profitable compared to simpler procedures. This is particularly true in lower-tier hospitals, where many patients cannot afford neurosurgical treatments. This financial barrier further contributes to a shortage in the demand for neurosurgical services.

In developed countries, neurosurgeons earn 5 to 6 times the average salary, while in developing countries like Saturn, the ratio is just 1 to 2 times, slightly above that of average workers. The

primary strategy to boost income is to focus exclusively on simpler but high-paying surgeries. These include procedures for superficial tumors such as scalp, skull, convex meningiomas, and pituitary adenomas, while avoiding more complex tumors like skull base or brainstem tumors. For instance, a neurosurgeon can earn 300 Satolar for excising a scalp lump in Saturn, whereas a microvascular decompression for hemifacial spasm yields only 800 Satolar. The benefits and risks of performing three scalp tumor excisions versus one microvascular decompression are clearly evident.

The fee-for-service system emphasizes "cost efficiency" over "efficiency." For example, while China has developed a similar DIP system to Saturn's, it features fewer diagnostic categories and less granularity than developed countries. Surgery on brain is distinct from orthopedics or plastic surgery, where treatment outcomes can often be quantified through imaging or aesthetic results. In contrast, neurological outcomes, such as impairments in memory or attention, are much more difficult to quantify. In systems with immature classification and evaluation methods, surgeries related to mental function should be approached with caution. These procedures not only carry complexity and legal risks, but they also face challenges in securing fair compensation.

Based on this understanding, surgeons may develop a "surgical tendency scale" (as shown in Figure 1) that assesses the potential benefits and risks of surgeries in their region. Surgical procedures will only be promoted when the benefits clearly outweigh the risks to the surgeon.

2. Embracing Selfishness

Every neurosurgeon has their own ethical guidelines. But how to define “ethics”? Is it the Hippocratic Oath that all physicians take, or the responsibilities they bear toward their families? Dagi and others have identified seven ethical issues in American healthcare reform such as physician accountability and data security(3), but the challenges in low-income countries are often more fundamental. Neurosurgeons in these regions face a survival crisis. While surgeons are accountable to their patients, this does not always

mean prioritizing their patient welfare over their families' interests. Altruism is considered a fundamental principle of medicine, but in harsh living conditions like Saturn, it may be a poison that leads to one's own downfall.

Low income lowers the ethical standards of neurosurgeons and incentivizes profit-driven behavior, such as accepting kickback. However, taking kickback is not a personal option, but a mandatory act seen as a “loyalty oath” in Saturn. As the Saturn government reduced funding for public hospital, it tacitly endorsed hospitals accepting kickbacks from pharmaceutical representatives. Kickbacks exist in every part of the system, including medicine approval, procurement, and prescription. One will be labeled a "heretic" if he or she is not part of the group. In the early phase of the fee-for-service system, the sale of pharmaceuticals and surgical products became one of the primary revenue streams for many public hospitals, a phenomenon referred to as “pharmaceuticals fund healthcare” (4). Kickbacks were gradually eliminated following the implementation of a centralized procurement policy.

Given that neurosurgeons undergo a much longer training period, many perceive themselves as being of a “higher class” than general physicians. Individuals from "higher social classes" tend to exhibit more selfish behavior and lower moral standards(5). Perhaps the economic inequality in Saturn renders unethical behavior more acceptable within the field of neurosurgery.

3. Collectivism Fosters Stronger Resilience

In a fee-for-service system, the government expects surgeons to operate like machines in a highly specialized factory. Unfortunately, surgeons are human beings with emotions, and expecting them to make every clinical decision as "rationally" as Dr. Fisher or Dr. Sand is a significant imposition.

Long-term ethical distress for surgeons may lead to burnout and mental diseases. However, socialist countries like Saturn emphasize collectivism, and physicians are more adept at enduring personal injustices, tending not to resist. Collectivism holds that "doing right is far more

important than doing good." This perspective often suppresses individuals who are brave enough to question and disclose injustices (Such as oncologist Zhang Yu(6)). As a result, the likelihood of developing mental health issues for physicians is lower in socialist countries compared capitalist countries. Statistics showed the physicians in China (0.14/100,000 per year(7)) has a significantly lower suicide number than American (5.82/100,000 per year(8)). Despite the strict control and oppression of individuals, the healthcare system reform in Saturn is making progress. This was largely attributed to the greater compliance of physicians in socialist countries.

4. Career development for Saturn's young neurosurgeon

The neurosurgeon's core competency is to achieve the best outcomes for patients, and the key lies in continual practice. In "Medicaid for All" system, resources and patients are predominantly concentrated in large cities and top-tier hospitals. Neurosurgeons in these hospitals can achieve more reasonable income by selecting patients. But in secondary hospitals, there is supply exceeding demand for neurosurgeons. Those surgeons have endless thirst for patients.

We find that this situation can be best understood through the lens of the "Offensive Realism" principle proposed by John Mearsheimer(9):

(1) Surgeons prioritize survival as their primary goal. Survival likelihood is positively correlated to their surgery volume.

(2) Each surgeon possesses specific techniques that are attractive to patients. Surgeons can dramatically increase their appeal to patients through innovative research or extensive advertising efforts.

(3) Surgeons are highly educated and rational actors, capable of devising effective strategies to maximize their chances of survival.

In such a Darwinian environment, neurosurgeons may face significant involution to maintain professional power and status. The surgery volume is the "absolute combat index", leading neurosurgeons to maximize their surgical volume at any cost, including purchasing search engine

traffic or even “buying” patients (as seen with Dr. Sand). They are less willing to share surgical techniques, leading to further decline in neurosurgery at secondary hospitals. Moreover, they may deliberately hide surgical details during teaching, causing their peers’ failures. In research area, overcompetition may foster a "publish or perish" mentality, where surgeons concentrate on generating research output or performing technically advanced surgeries. They may pursue new technologies solely to stay ahead of their competitors, regardless of whether the innovation represent the best options for patient care (e.g., CRISPR babies(10)).

Life in Saturn’s isn’t easy. For young neurosurgeons, the most practical way to escape low salaries is to stockpile surgery volume and fame in public hospital by fair means or foul, ultimately aspiring to become renowned surgeons and perform high-reward surgeries in private hospitals.

IV. Conclusions

The practice of two neurosurgeons on Saturn offers a unique opportunity to observe healthcare reform in a socialist system, which is a minority social ideology on Earth. Many socialist countries are committed to universal healthcare, characterized by low costs and equitable access to services. However, this may bring both economic and ethical challenges for high-cost fields such as neurosurgery. Survival stress on Saturn might compel neurosurgeons to engage in various unethical practices that violate the Hippocratic Oath they once swore.

V. Materials and Methods

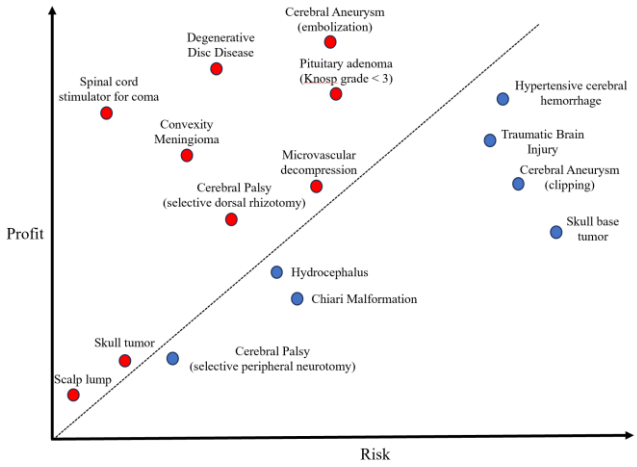

Figure 1. Saturn neurosurgery surgical tendency scale provided by Dr. Fisher, in the Saturn TO38

year, following the reforms in the "work points system" and "annual salary system".

Note that the profit and risk of diseases/surgery may vary depending on the region and policies. RED represents recommended surgeries where the profit exceeds the risk, while BLUE represents unrecommended surgeries where the risk exceeds the profit.

Supplementary Materials

NA

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